

Heesung Han

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EDUCATION

Olin College of Engineering, Needham, MA | GPA: 4.0/4.0

Expected May 2028

Bachelor of Science, Mechanical Engineering

- Relevant coursework: Multivariable Calculus, Linear Algebra, Thermodynamics, Software Design, Principles of Integrated Engineering

California Academy Of Math and Science, Carson, CA

June, 2024

- Relevant coursework: Engineering Design and Development, Computer-Integrated Manufacturing

EXPERIENCE

Olin Baja Racing Team (Baja SAE), Needham, MA

Aug, 2024 - Present

Drivetrain Subteam Member

- Managed the Rear Output by designing and manufacturing components, including the rear spool, sprockets, bearing mounts, and brake rotors
- Collaborated with a cross-functional team to improve drivetrain efficiency and vehicle reliability
- Tested and iterated on mechanical assemblies to collect data on torque output

Olin Robotics Lab, Needham, MA

Aug, 2024 - Present

Agricultural Robotics Student Researcher

- Collaborating with an MIT PhD candidate to develop a soil moisture sampling drone for precision agriculture
- Designed a drone docking station that autonomously opens and closes based on the drone's status
- Implemented a one-axis gantry system for a weed-picking robot

Simplehuman, Torrance, CA

July, 2024 - Aug, 2024

Engineering Intern (Research and Development)

- Created prototypes and mockups to test the consistency and performance of home appliance product designs using Autodesk Fusion 360, 3D printing, and laser cutting
- Integrated and assessed various sensors of the prototype to enhance functionality and reliability
- Communicated with electrical, finance, systems, and mechanical subteams to develop potential products

First Robotics Competition (FRC) 687 Nerd Herd, Carson, CA

May, 2023 - June, 2024

Captain

- Led 100+ members, particularly in the CAD and Strategy subteams, to win the FRC World Championship in Texas
- Taught CAD members and leaders efficient methods for designing robots using Autodesk Fusion 360
- Strengthened communication with mentors and members through weekly meetings and progress tracking

PROJECTS

Box Bot (Course: Principles of Integrated Engineering)

October, 2025 - December, 2025

- Designed and built a box-moving robot capable of transporting loads up to 15 lb and boxes up to 28.5 in
- Led a 4-person team as Project Manager using Gantt charts, BOMs, and bi-weekly sprints to keep track

Face Detection Video Game (Course: Software Design)

March, 2025 - May, 2025

- Developed a video game controlled by real-time facial expressions using Python and OpenCV

Autonomous Wall-Climbing Robot (Course: Engineering Design and Development)

Aug, 2023 - June, 2024

- Built a propeller-driven robot capable of generating thrust to adhere to vertical surfaces

SKILLS

Software: Autodesk Fusion 360, STK, Onshape, SOLIDWORKS, MATLAB, Python

Fabrication: 3D Printing, Laser Cutting, Bandsaw, Drill Press, CNC Mill Machine, CNC Lathe Machine

Languages: English (fluent), Korean (conversational)